Test Report - VxMark Backup Power

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Summary

Testing showed that the 1500VA/1000W UPS can power the VxMark unit and printer for more than 2 hours.

Applicable VVSG Requirements

- 2.7-H – Power outages, sags, and swells

Devices Under Test

- VxMark, v4.1, image v4.0.4
- HP4001dn printer
- Goldenmate 1500VA/1000W UPS

Purpose

VxMark must continue functioning on backup power for at least 2 hours in case of a power outage, to satisfy requirement 2.7-H in VVSG2.0. This report documents tests the Goldenmate 1500VA/1000W UPS to keep powering the VxMark unit and HP4001dn printer over time when disconnected from mains power.

Materials

- VxMark unit (MK-14-019)
    - Imaged with v4.0.4 production app
    - Configured with summary ballot test election
    - Connected to printer
- VxMark printer (HP4001dn, VNL0728678), paper
- Goldenmate 1500VA/1000W UPS (JUO2HE390707)
- Timer

Procedures

1. Charge the UPS to 100% overnight plugged into mains power.
2. Set up the VxMark system, with the VxMark unit and HP printer plugged into the UPS.
3. Unplug the UPS from mains power and start a timer.
4. Regularly check the system power.
5. Every 10-30 minutes, print out batches of ballots, to total at least 50 across 2 hours.

Results

The UPS continued powering the system without issue for over 2 hours and 21 minutes, going from 100% charge to 60% charge in that time. The VxMark system printed 49 summary ballots in 2 hours, and 51 ballots in the full 2 hours and 21 minutes tested. Testing was done at room temperature indoors.

Conclusions

- The Goldenmate 1500VA/1000W UPS can power the VxMark system for at least 2 hours when unplugged from mains if charged properly beforehand.
- This is when printing summary ballots at a rate of approximately 2.4 minutes per ballot. A real election is expected to print at a slower rate than this and can have even less charge used over time than seen here.

Appendix A: VxMark thermal measurements

Thermals with ADATA SSD, updated hardware and software March 2026				**Temperature measurements (degrees F)**					
				Room temperature ambient		**Warm ambient**			
Location	**User-facing?**	**Type**	**Measurement tool**	**Idle 30+ min**	**Idle overnight**	**On 2 hours, 95F, 55%RH**	**On 3 hours, 95F, 55%RH**	**On 4 hours, 95F, 55%RH**	**Spec: Max operating (F)**
right plastic panel above SBC, under ATI	yes	surface	thermal imager	101.7	100.6	119.5	122.8	117.2	140 (OSHA)
USB port	yes	surface	thermal imager	97.8	92.5	115.0	119.5	115.0	140 (OSHA)
rear plastic cover where cords exit	yes	surface	thermal imager	84.5	87.0	110.1	111.2	106.2	140 (OSHA)
left horizontal plastic panel rear to headphone storage	yes	surface	thermal imager	85.5	85.6	108.6	110.4	105.8	140 (OSHA)
Barcode scanner light	yes	surface	thermal imager	92.2	93.3	116.5	119.2	116.1	140 (OSHA)
Touchscreen, center	yes	surface	thermal imager	82.1	81.6	105.4	105.4	100.4	140 (OSHA)
SSD chip	no	surface	thermal imager	129.6	not measured	161.7	154.7	149.3	n/a
Power supply internals	no	surface	thermal imager	122.9	not measured	146.1	145.6	141.9	158 (spec)
SBC front by bracket, chip	no	surface	thermal imager	118.5	not measured	151.1	144.9	139.5	n/a
barcode scanner rear	no	surface	thermal imager	98.6	not measured	131.4	125.9	121.5	140F (spec)
SSD chip, ambient	no	air, surrounding	thermocouple	not measured		111.1	not measured		185 (spec)
SBC front by bracket, ambient	no	air, surrounding	thermocouple	not measured		114.2	117.1	116.8	140F (spec)